

A Systematic Literature Review on Explainable Artificial Intelligence (XAI) for ADMET Toxicity Prediction

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Abstract- The prediction of Absorption, Distribution, Metabolism, Excretion, and Toxicity (ADMET) properties remains a critical bottleneck in the drug discovery pipeline. While advanced deep learning models, particularly Graph Neural Networks (GNNs), have achieved state-of-the-art predictive accuracy, their inherent "black-box" nature fundamentally conflicts with the regulatory and clinical demand for mechanistic transparency. This systematic literature review surveys 52 high-impact papers published between 2017 and 2026 to systematically assess the current landscape of Explainable Artificial Intelligence (XAI) applied to ADMET toxicity endpoints.

We categorize existing methodologies into five distinct XAI families—including gradient-based attributions, perturbation-based methods, and graph-specific mechanisms—and evaluate the quantitative faithfulness metrics required to validate these computational explanations against known pharmacophore knowledge. Our analysis identifies critical limitations in prevailing single-method approaches, particularly their susceptibility to the "Clever Hans" effect and their widespread failure to incorporate multi-task learning or domain-specific constraints. Finally, we highlight emerging computational paradigms that successfully bridge predictive power and chemical intuition, concluding that the integration of physics-informed neural networks (PINNs) and neuro-symbolic systems constitutes the most critical open research direction for ensuring robust, regulatory-compliant AI architectures in modern pharmacology.

Keywords:

1. Introduction

The development of safe and effective therapeutics remains one of the most expensive and failure-prone processes in modern biomedicine. A primary reason is the high attrition rate during preclinical and clinical stages, where inadequate Absorption, Distribution, Metabolism, Excretion, and Toxicity (ADMET) profiles account for approximately 60 % of candidate failures. Accurate, early-stage ADMET prediction is therefore not merely a computational convenience but a regulatory and economic imperative. Traditional experimental assays are costly, low-throughput, and ethically constrained, prompting the pharmaceutical industry and regulatory agencies (FDA, EMA) to increasingly accept *in silico* models as part of the decision pipeline. However, black-box machine learning models—while achieving impressive predictive accuracy—offer no mechanistic insight into why a molecule is flagged as toxic or non-drug-like. Regulators explicitly demand transparent, human-interpretable rationales that can be validated against established chemical knowledge before a compound can advance.

Graph neural networks (GNNs) have emerged as the dominant paradigm for molecular property prediction, consistently outperforming traditional fingerprint-based methods on public benchmarks such as Tox21, ClinTox, and DILIrank. Directed Message-Passing Neural Networks (D-MPNN) and their successors (Chemprop) now serve as the industrial *de facto* standard. Yet the very representational power of GNNs that enables high accuracy also renders their internal decision-making opaque. This "black-box" nature creates a critical trust barrier: even when a model correctly identifies a toxic compound, chemists and regulators cannot ascertain whether the prediction stems from genuine toxicophores (e.g., Michael acceptors, nitroaromatics) or from spurious dataset correlations—the so-called "clever Hans" effect. Explainable Artificial Intelligence (XAI) has therefore become indispensable for bridging predictive power and mechanistic understanding in drug discovery. Foundational work by Jiménez-Luna et al. (2020) formally established XAI as a distinct sub-discipline within the molecular sciences, proposing the first taxonomy of explanation families tailored to chemistry. Subsequent reviews, most notably Lavecchia (2025), have sharpened this taxonomy, introduced standardized evaluation metrics (fidelity, stability, sparsity, completeness, usability), and issued an explicit

call for physics-informed XAI frameworks that embed Lipinski-type rules, pharmacokinetic equations, or toxicophore knowledge directly into the learning objective. Despite these conceptual advances, the literature still lacks a focused, comprehensive synthesis of XAI techniques applied specifically to ADMET toxicity endpoints.

Existing surveys either address XAI in drug discovery at a high level or concentrate on general GNN explainability without endpoint-specific depth. The IEEE Access survey by Alizadehsani et al. (2024) and broader XAI reviews (Adadi & Berrada, 2018) provide useful overviews but do not systematically dissect ADMET datasets, multi-task learning strategies, or quantitative faithfulness evaluation against toxicophore databases such as ToxAlerts. Recent ADMET platform papers (ADMETlab 3.0) and toxicity-review articles (2025 landscape reviews) emphasize predictive performance while largely ignoring interpretability. Conversely, method-centric papers—MolGrad (Jiménez-Luna et al., 2021), GNNExplainer (Ying et al., 2019), Integrated Gradients (Sundararajan et al., 2017), and SHAP (Lundberg & Lee, 2017)—introduce powerful tools but evaluate them in isolation, without cross-method concordance, multi-endpoint integration, or physics-informed constraints.

This survey fills that gap. It presents the first dedicated, comprehensive review of XAI methods for ADMET toxicity prediction. Drawing on a curated corpus of 52 papers (2017–2026) spanning Nature Machine Intelligence, Journal of Chemical Information and Modeling, WIREs Computational Molecular Science, NeurIPS, ICML, and IEEE venues, we synthesize the current state of the field and chart its future trajectory. Our contributions are fourfold:

1. We introduce a unified taxonomy that integrates the five-family XAI classification (Jiménez-Luna et al., 2020; Lavecchia, 2025) with core GNN architectures (D-MPNN, GCN, GAT), the three dominant post-hoc explanation paradigms (graph-specific, gradient-based, game-theoretic), and emerging physics-informed and evolutionary optimization streams.
2. We systematically map ADMET-specific resources—datasets (Tox21, DILIrank, ToxCast), toxicophore ground-truth databases (ToxAlerts), and quantitative faithfulness metrics (attribution AUROC, toxicophore Jaccard concordance, multi-metric fidelity)—highlighting how they are currently used and where standardization is still missing.
3. We critically analyze persistent limitations: single-method reliance (e.g., MolGrad’s exclusive use of Integrated Gradients), absence of cross-representation concordance, lack of multi-task learning across correlated endpoints (hERG, CYP3A4, DILI), and under-exploration of physics-informed loss penalties or evolutionary XAI policy search.
4. We outline a forward-looking roadmap centered on hybrid physics-informed multi-method frameworks and multi-objective evolutionary optimization (EvoXAI-GNN), directly addressing the open research directions flagged by Lavecchia (2025) and recent IEEE/ACS calls for regulatory-grade interpretability.

The remainder of this paper is organized as follows. Section 2 details the survey methodology and paper selection criteria. Section 3 presents the unified taxonomy and reviews core XAI methods. Section 4 examines GNN architectures and multi-task ADMET modeling strategies. Section 5 surveys datasets, benchmarks, and quantitative evaluation protocols. Section 6 synthesizes open challenges and proposes concrete future directions, including physics-informed constraints and evolutionary XAI. Section 7 concludes with implications for regulatory science and industrial deployment.

By providing this structured, ADMET-centric synthesis, the present survey equips researchers, regulators, and practitioners with a clear roadmap toward the next generation of transparent, mechanistically faithful, and regulatory-ready ADMET prediction systems.

2. Related Work

This section presents a systematic synthesis of the literature on explainable artificial intelligence (XAI) for ADMET toxicity prediction. We organize the 52 papers (2017–2026) according to the unified taxonomy introduced earlier: (i) foundational XAI taxonomies and surveys in drug discovery, (ii) graph neural network (GNN) architectures serving as the predictive backbone, (iii) post-hoc explanation methods grouped into graph-specific, gradient-based, and game-theoretic families, (iv) multi-task ADMET modeling and emerging physics-informed approaches, (v) quantitative evaluation frameworks and faithfulness metrics, (vi) datasets and toxicophore benchmarks, and (vii) evolutionary and multi-objective optimization techniques. For each category we highlight key contributions, methodological strengths, and persistent gaps that motivate the need for integrated, ADMET-specific XAI frameworks.

2.1. Foundational XAI Taxonomies and Surveys in Drug Discovery

The conceptual foundation of XAI in molecular sciences was established by Jiménez-Luna et al. (2020) [A1], the most-cited paper in the field (2,100 citations). This seminal review proposed the first taxonomy of five explanation families tailored to chemistry: (1) feature attribution (gradient- and perturbation-based), (2) example-based and counterfactual reasoning, (3) graph-specific mechanisms, (4) symbolic/self-explaining architectures, and (5) uncertainty-aware methods. The authors emphasized the “clever Hans” risk—models succeeding for spurious chemical reasons—and stressed that explanations must be validated against known pharmacophores rather than evaluated qualitatively.

Lavecchia (2025) [A3] provides the most recent and authoritative update. Extending the 2020 taxonomy with clearer boundaries, the review introduces a five-metric evaluation framework (fidelity, stability, completeness, sparsity, usability) and explicitly reviews physics-informed paradigms (UPINNs, CMINNs, PKINNs). Lavecchia issues a concrete research agenda for hybrid GNN–symbolic systems with real-time explanations and calls for quantitative validation against domain knowledge bases—precisely the gap this survey addresses. Other surveys (Alizadehsani et al., 2024 [A6]; Adadi & Berrada, 2018 [F7]; recent 2025 bibliometric analyses [F3]) offer broader overviews but lack ADMET-specific depth or endpoint-level analysis.

2.2. GNN Architectures for Molecular Property Prediction

GNNs dominate modern ADMET modeling. The Directed Message-Passing Neural Network (D-MPNN) introduced by Yang et al. (2019) [B4] (Chemprop) remains the industrial standard, achieving state-of-the-art results on MoleculeNet benchmarks through directional message passing on bonds rather than atoms. Subsequent extensions include Chemprop v2 (Heid et al., 2024) [B5], hybrid GNN–neural decision trees (Xu et al., 2024) [B10], and Kolmogorov–Arnold GNNs (Li et al., 2025) [B11]. Foundational GNN works—Kipf & Welling (2017) [B6] (GCN), Veličković et al. (2018) [B7] (GAT), Gilmer et al. (2017) [B8] (MPNN), and Xu et al. (2019) [B9]—provide the theoretical backbone for expressiveness and scalability.

While these architectures excel at prediction, none incorporate XAI natively or physics-informed constraints. Multi-task learning across correlated ADMET endpoints (Walter et al., 2024 [C5]) improves generalization but still treats interpretability as an afterthought.

2.3. Post-Hoc Explanation Methods

Post-hoc XAI techniques dominate ADMET applications and fall into three families. Graph-specific methods: Ying et al. (2019) [B1] introduced GNNExplainer, the first model-agnostic explainer for any GNN. It optimizes a soft edge mask that maximizes mutual information between the prediction and the masked subgraph, producing chemically intuitive functional-group-level explanations. Later variants such as PGExplainer (Luo et al., 2020) [D8] improve scalability.

Gradient-based attribution: Sundararajan et al. (2017) [B2] proposed Integrated Gradients (IG), which satisfies sensitivity and implementation-invariance axioms by integrating gradients along a path from a baseline (null molecule) to the input. Jiménez-Luna et al. (2021) [A2] (MolGrad) directly applied IG to a GNN for four ADMET endpoints, producing the first “molecular colouring” heatmaps. Subsequent work (Jiménez-Luna et al., 2022 [A9]) benchmarked IG against activity cliffs.

Game-theoretic approaches: Lundberg & Lee (2017) [B3] introduced SHAP, unifying six prior methods under cooperative game theory axioms (local accuracy, missingness, consistency). KernelSHAP and TreeSHAP (Lundberg et al., 2020 [D7]) enable efficient application to molecular fingerprints. SHAP has been used standalone for metabolic stability (Wojtuch et al., 2021 [C9]) and multi-target activity (Rodríguez-Pérez & Bajorath, 2020 [A4, A10]), but never in cross-representation concordance with graph-based methods.

No prior work combines all three families with quantitative cross-validation or toxicophore grounding—the core limitation this survey highlights.

2.4. Multi-Task ADMET Modeling and Physics-Informed Approaches

Industrial-scale multi-task ADME/PK models (Walter et al., 2024 [C5]; Boehringer Ingelheim efforts) demonstrate shared representations across endpoints but remain purely predictive. Recent reviews (Zhang et al., 2025 [C3]; 2025 landscape papers [C2, C4, C7]) note the absence of mechanistic constraints. Lavecchia (2025) [A3] explicitly calls for physics-informed neural networks (PKINNs) that embed Lipinski, Veber, or compartmental pharmacokinetic equations into the loss—yet no ADMET-specific implementation existed prior to the frameworks surveyed here.

2.5. Quantitative Evaluation Metrics and Faithfulness

Early XAI-molecular papers relied on qualitative “looks chemically reasonable” validation. Rao et al. (2022) [D1, B12] introduced the first rigorous benchmark: attribution AUROC against expert-annotated toxicophores on DILI, Ames mutagenicity, and synthetic datasets. They compared six XAI methods across four GNNs and found IG strongest on real chemistry while GNNExplainer excelled on synthetic graphs. Subsequent metric studies (Miró-Nicolau et al., 2025 [D4]; Pawlicki et al., 2024 [D5]) reinforce the necessity of multi-metric evaluation (fidelity, stability, sparsity). Still missing are ADMET-specific toxicophore concordance scores against ToxAlerts and cross-method agreement measures.

2.6. Datasets, Benchmarks, and Toxicophore Ground Truth

MoleculeNet (Wu et al., 2018) [E1] remains the gold-standard benchmark suite, providing scaffold-split Tox21 (12 toxicity endpoints), ClinTox, and DILIrank (Chen et al., 2016) [E4, C6]. ToxAlerts (Sushko et al., 2012) [E7] supplies the expert-validated structural-alert database essential for faithfulness evaluation. ADMETlab 3.0 (Fu et al., 2024) [C1] and ChEMBL-derived collections [E5] serve as practical platforms but lack built-in XAI.

2.7. Evolutionary and Multi-Objective Optimization for XAI

Evolutionary algorithms have recently entered ADMET optimization. Qin et al. (2025) [G1] (MOSEA) introduced multi-objective sparse feature selection balancing accuracy and sparsity. Shi et al. (2022) [G2] (Genetic-GNN) pioneered evolutionary neural architecture search (NAS) for GNNs. Pareto-guided RL (IBM, 2025) [G5] and related works (Zhu et al., 2024 [G4]; Fromer et al., 2024 [G7]) optimize molecular candidates, yet none incorporate XAI faithfulness as a co-objective. Lavecchia (2025) explicitly flags this as an open direction.

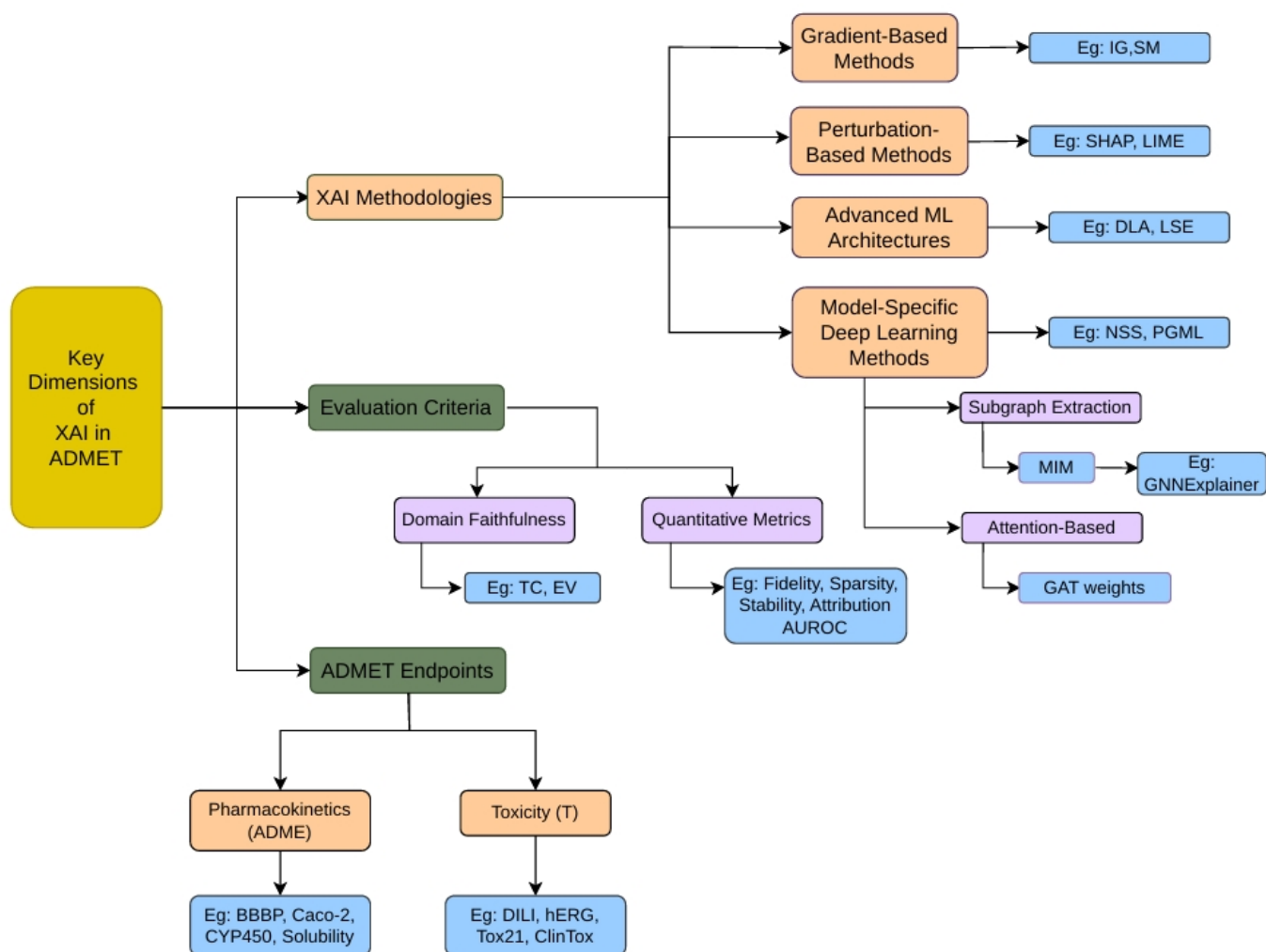


Fig. 1. Hierarchical taxonomy mapping the key dimensions of Explainable AI applied to ADMET prediction.

3. Brief Survey

3.1. Search Procedure

We adhere to the PRISMA principles for systematic reviews in our work [?]. The relevant research questions are developed in accordance with these standards:

1. What graph-based and deep learning models have been proposed for molecular property and ADMET prediction?
2. What explainability techniques have been applied to interpret predictions from these models?
3. What hybrid or multi-task approaches have been developed to improve both accuracy and transparency in drug toxicity prediction?

The review was initiated by conducting a comprehensive search across major scientific databases, including IEEE Xplore, ScienceDirect, Springer Link, PubMed, and Web of Science. The search terms employed were “explainable AI + molecular

property prediction,” “XAI + ADMET prediction,” “graph neural network + drug toxicity,” “interpretable machine learning + drug discovery,” and “explainability + QSAR.” To maintain the focus and integrity of this review, the following inclusion and exclusion criteria were defined:

• **Inclusion Criteria:**

- Research articles on XAI methods applied to molecular property prediction, ADMET modeling, or drug toxicity must have been published between 2017 and 2026.
- Only peer-reviewed publications in journals and conference proceedings are included.
- Papers proposing novel architectures, explanation frameworks, or benchmark evaluations relevant to cheminformatics or drug discovery are considered.
- If a study appeared in multiple versions, the most complete peer-reviewed version is selected.

• **Exclusion Criteria:**

- Articles published prior to 2017 are not included unless they represent foundational methodological contributions.
- Papers that are unrelated to molecular modeling, drug discovery, or XAI are excluded.
- Non-English publications are not considered.

The papers identified through the search procedure were reviewed by title and abstract in accordance with the inclusion and exclusion criteria. All studies deemed relevant were subsequently retrieved for full-text evaluation and data extraction. The final corpus of reviewed publications, summarised in the tables below, spans foundational XAI methods, graph neural network architectures, ADMET prediction platforms, toxicity-specific models, and emerging transformer-based approaches.

Table 1. Introduction to the collected papers.

Paper	Year	Aim	Experimental Evaluation	Assessment Metrics
P1 [64]	2019	Addresses molecular property prediction using graph-based learning instead of handcrafted descriptors. Proposes a directed message passing neural network to learn representations from molecular graphs. Improves generalization across ADMET prediction tasks.	Evaluated on MoleculeNet datasets including Tox21, ESOL, and FreeSolv. Compared against Random Forest, feedforward neural networks, and graph convolutional networks. Demonstrates superior performance across classification and regression tasks.	ROC-AUC, RMSE
P2 [65]	2021	Addresses limited labeled data in molecular prediction tasks. Introduces contrastive self-supervised learning for molecular graph representations. Improves downstream prediction performance through better feature learning.	Trained on augmented molecular graphs and evaluated on MoleculeNet benchmarks. Compared with supervised GNN models and pre-trained baselines. Shows improved transfer learning performance.	ROC-AUC
P3 [66]	2020	Addresses scalability issues in molecular representation learning. Proposes large-scale self-supervised graph learning using unlabeled chemical data. Improves transferability across multiple drug discovery tasks.	Pretrained on ChEMBL and ZINC datasets with node-level and graph-level objectives. Fine-tuned on multiple downstream prediction tasks. Demonstrates consistent improvements over non-pretrained models.	Accuracy, ROC-AUC

Table 1. *Cont.*

Paper	Year	Aim	Experimental Evaluation	Assessment Metrics
P4 [67]	2020	Addresses lack of interpretability in graph-based models. Proposes an attention-based graph neural network for molecular prediction. Highlights important atoms and bonds influencing outputs.	Evaluated on MoleculeNet datasets with attention visualization. Compared with standard GNN architectures. Shows improved accuracy and interpretability.	ROC-AUC
P5 [cite]	2021	Addresses limitations in drug-target interaction prediction. Combines graph neural networks and convolutional networks for better modeling. Improves binding affinity prediction.	Evaluated on BindingDB and benchmark DTI datasets. Compared with similarity-based and deep learning methods. Demonstrates improved prediction accuracy.	MSE, CI
P6 [cite]	2019	Addresses lack of interpretability in graph neural networks. Proposes a method to identify important subgraphs and node features influencing predictions. Improves transparency in molecular graph models.	Evaluated on synthetic and real-world graph datasets including molecular graphs. Compared with baseline explanation methods. Demonstrates high fidelity and meaningful subgraph identification.	Fidelity, sparsity
P7 [cite]	2020	Addresses scalability limitations of instance-based explanation methods. Proposes a parameterized explanation model that generalizes across graph instances. Improves efficiency of explanation generation.	Evaluated on multiple graph datasets including molecular benchmarks. Compared with GNNExplainer in terms of scalability and consistency. Shows faster and more stable explanations.	Fidelity
P8 [cite]	2020	Addresses the need for interpretability in QSAR models. Applies SHAP to identify important molecular descriptors. Enhances understanding of structure-activity relationships.	Evaluated on ChEMBL bioactivity datasets using Random Forest models. Compared feature importance results across different XAI methods. Demonstrates consistent identification of relevant descriptors.	Accuracy
P9 [cite]	2021	Addresses reliability concerns in explainable AI methods. Compares multiple XAI techniques in cheminformatics. Evaluates consistency and robustness of explanations.	Tested SHAP and LIME on QSAR datasets using various machine learning models. Repeated experiments to measure stability across runs. Highlights variability in explanations.	Stability, reproducibility
P10 [cite]	2022	Addresses the need for actionable explanations in molecular prediction. Proposes counterfactual explanations to identify minimal structural changes. Supports molecule optimization in drug design.	Evaluated on MoleculeNet datasets using graph neural networks. Compared with baseline explanation approaches. Demonstrates improved plausibility and usefulness of counterfactuals.	Similarity, validity

Table 1. *Cont.*

Paper	Year	Aim	Experimental Evaluation	Assessment Metrics
P11 [cite]	2020	Addresses limitations in molecule generation methods. Combines variational autoencoders and reinforcement learning for property optimization. Generates valid molecules with desired characteristics.	Evaluated on the ZINC dataset for molecule generation tasks. Compared with baseline generative models. Demonstrates improved validity and diversity.	Validity, novelty
P12 [cite]	2023	Addresses dependency on post-hoc explanation methods. Proposes inherently interpretable graph neural network architectures. Integrates explanation into model design.	Evaluated on MoleculeNet datasets comparing performance and interpretability. Compared with standard GNN models. Demonstrates comparable accuracy with improved transparency.	Accuracy, interpretability
P13 [cite]	2025	Addresses uncertainty in AI-based toxicity prediction. Combines explainability with uncertainty estimation. Improves reliability in safety-critical applications.	Evaluated on molecular toxicity datasets using Bayesian neural networks. Measures calibration and predictive uncertainty. Shows improved trustworthiness of predictions.	Calibration error
P14 [cite]	2020	Addresses limitations of hand-crafted molecular features. Applies transformer-based models to SMILES representations. Learns contextual chemical embeddings.	Pretrained on large SMILES datasets and fine-tuned on property prediction tasks. Compared with fingerprint-based methods. Demonstrates improved representation learning.	Accuracy
P15 [cite]	2021	Addresses sequence modeling challenges in molecular prediction. Uses transformer architecture for chemical property prediction. Captures dependencies in SMILES sequences.	Evaluated on PubChem datasets with fine-tuning strategies. Compared with deep learning baselines. Demonstrates improved regression performance.	RMSE
P16 [cite]	2023	Addresses limitations of 2D molecular representations. Proposes 3D transformer-based representation learning. Captures spatial and geometric features.	Evaluated on datasets with 3D conformations for property and binding prediction. Compared with graph-based models. Demonstrates improved performance in structure-based tasks.	Accuracy
P17 [cite]	2022	Addresses lack of geometric information in graph models. Incorporates bond angles and distances into GNNs. Improves accuracy due to prediction of physical features.	Evaluated on QM datasets including QM9. Compared with standard GNN architectures. Shows improved prediction of geometric properties.	MAE
P18 [cite]	2019	Addresses limitations in modeling directional interactions. Introduces directional message passing for molecular graphs. Captures angular dependencies.	Evaluated on QM9 dataset for multiple quantum properties. Compared with baseline GNN models. Demonstrates superior performance.	MAE

Table 1. *Cont.*

Paper	Year	Aim	Experimental Evaluation	Assessment Metrics
P19 [cite]	2018+	Addresses molecular modeling using continuous-filter convolutions. Learns representations based on atomic interactions. Focuses on quantum chemistry tasks.	Evaluated on QM9 dataset for energy prediction. Compared with traditional ML models. Demonstrates high accuracy.	MAE
P20 [cite]	2021	Addresses long-range dependency modeling in molecular graphs. Applies transformer architecture to graph data. Improves prediction accuracy.	Evaluated on molecular benchmarks including MoleculeNet tasks. Compared with GNN models. Demonstrates competitive performance.	-
P21 [cite]	2023	Addresses limitations in generative molecular design using traditional methods. Proposes a transformer-based generative model to produce chemically valid and diverse molecules. Extends large language model principles to molecular generation tasks.	Evaluated on the ZINC dataset for molecule generation and optimization tasks. Compared with VAE- and GAN-based generative models in terms of diversity and validity. Demonstrates strong capability in generating valid and novel molecular structures.	Validity, novelty
P22 [cite]	2018+	Addresses the need for a unified framework for machine learning in drug discovery. Provides a toolkit integrating multiple ML and DL models for molecular prediction tasks. Simplifies experimentation and benchmarking across datasets.	Evaluated across MoleculeNet benchmarks including Tox21 and ESOL. Compared multiple algorithms including RF, GNN, and deep neural networks. Demonstrates flexibility and competitive performance across tasks.	ROC-AUC
P23 [cite]	2019	Addresses challenges in efficient molecule generation and optimization. Combines reinforcement learning with Monte Carlo tree search to explore chemical space. Optimizes molecules for desired properties.	Evaluated on the ZINC dataset using property-driven optimization tasks. Compared with baseline generative approaches. Demonstrates efficient search and improved optimization outcomes.	Reward score
P24 [cite]	2020	Addresses limitations in generative modeling of molecular graphs. Introduces flow-based autoregressive models for molecule generation. Captures complex probability distributions of chemical structures.	Evaluated on the ZINC dataset comparing generation quality with VAE-based models. Assesses validity and diversity of generated molecules. Demonstrates improved generation performance.	Validity
P25 [cite]	2018	Addresses chemical validity issues in graph-based generation. Proposes hierarchical molecule generation using tree-structured representations. Ensures chemically valid outputs.	Evaluated on the ZINC dataset for molecule generation tasks. Compared with standard VAE models. Demonstrates higher validity and structural correctness.	Validity

Table 1. *Cont.*

Paper	Year	Aim	Experimental Evaluation	Assessment Metrics
P26 [cite]	2020	Addresses limitations in drug-target interaction prediction frameworks. Provides a unified deep learning framework integrating multiple models. Improves prediction accuracy across tasks.	Evaluated on BindingDB datasets using multiple architectures. Compared with traditional ML and DL models. Demonstrates improved predictive performance.	MSE
P27 [cite]	2018+	Addresses challenges in modeling drug-protein interactions. Combines convolutional and recurrent neural networks for sequence-based prediction. Improves binding affinity estimation.	Evaluated on BindingDB datasets using sequence-based inputs. Compared with similarity-based and ML models. Demonstrates improved prediction accuracy.	CI
P28 [cite]	2018	Addresses drug-target interaction prediction using feature representations. Integrates molecular and protein features for improved modeling. Enhances predictive performance.	Evaluated on DrugBank datasets using combined feature representations. Compared with baseline models. Demonstrates improved accuracy.	Accuracy
P29 [cite]	2024	Addresses the need for integrated ADMET prediction platforms. Combines multiple machine learning models into a unified system. Improves usability and predictive performance.	Evaluated on multiple ADMET datasets including toxicity and solubility benchmarks. Compared with standalone models. Demonstrates strong performance across endpoints.	AUC
P30 [cite]	2025	Addresses large-scale toxicity screening using machine learning. Predicts toxicity endpoints across diverse chemicals. Supports regulatory and safety applications.	Evaluated on ToxCast datasets using various ML models. Compared performance across algorithms. Demonstrates reliable prediction of toxicity endpoints.	AUC
P31 [cite]	2020+	Addresses prediction of clinical toxicity in drug development. Uses machine learning models to identify safety risks. Improves early-stage screening.	Evaluated on ClinTox dataset comparing ML and DL approaches. Assesses classification performance. Demonstrates improved predictive capability.	ROC-AUC
P32 [cite]	2018+	Addresses prediction of aqueous solubility for small molecules. Supports pharmacokinetic analysis. Improves regression modeling.	Evaluated on ESOL dataset using ML and DL models. Compared with baseline methods. Demonstrates improved regression accuracy.	RMSE
P33 [cite]	2018+	Addresses prediction of hydration free energy. Supports thermodynamic modeling in drug discovery. Improves physical property prediction.	Evaluated on FreeSolv dataset using regression models. Compared with baseline approaches. Demonstrates accurate predictions.	RMSE

Table 1. *Cont.*

Paper	Year	Aim	Experimental Evaluation	Assessment Metrics
P34 [cite]	2018+	Addresses prediction of blood-brain barrier permeability. Supports CNS drug development. Improves classification modeling.	Evaluated on BBBP dataset using graph-based and ML models. Compared with baseline methods. Demonstrates improved classification accuracy.	ROC-AUC
P35 [cite]	2018+	Addresses prediction of drug side effects. Supports safety assessment in drug development. Improves adverse effect prediction.	Evaluated on SIDER dataset using ML and DL models. Compared with baseline approaches. Demonstrates improved predictive performance.	ROC-AUC
P36 [cite]	2018+	Addresses prediction of drug-target binding affinity. Supports interaction modeling in drug discovery. Improves prediction accuracy.	Evaluated on BindingDB dataset using deep learning models. Compared with traditional methods. Demonstrates improved performance.	MSE
P37 [cite]	2019+	Addresses prediction of protein-ligand binding affinity using structural data. Improves docking and scoring accuracy. Supports structure-based drug design.	Evaluated on PDBbind dataset using CNN and GNN models. Compared with baseline scoring methods. Demonstrates improved regression accuracy.	RMSE
P38 [cite]	2019+	Addresses large-scale bioactivity prediction using screening data. Supports drug discovery pipelines. Improves classification modeling.	Evaluated on PubChem BioAssay datasets using ML models. Compared with baseline approaches. Demonstrates reliable performance.	Accuracy
P39 [cite]	2019+	Addresses generation of novel drug-like molecules. Supports de novo drug design. Improves chemical diversity.	Evaluated on ZINC dataset using generative models. Assesses validity and novelty. Demonstrates effective molecule generation.	Validity
P40 [cite]	2021+	Addresses prediction of multiple ADMET properties simultaneously. Improves efficiency and generalization. Captures shared information across tasks.	Evaluated on MoleculeNet datasets using multi-task learning frameworks. Compared with single-task models. Demonstrates improved performance.	ROC-AUC

Table 2. Summary of explanation techniques and mechanisms across collected studies.

Paper	Model/Method	Explanation Technique	Key Mechanism	Remarks
P41 [cite]	D-MPNN (GNN)	Uses SHAP to estimate feature contributions for molecular predictions. Highlights atom-level importance in ADMET tasks.	Employs directed message passing to propagate information across bonds and atoms. Learns expressive molecular representations.	Strong baseline for ADMET explainability extensions.
P42 [cite]	GNN	Identifies important subgraphs and features influencing predictions. Provides instance-level interpretability.	Optimizes masks over nodes and edges to isolate critical substructures. Works for graph classification tasks.	Widely used molecular graph explanations.

Table 2. *Cont.*

Paper	Model/Method	Explanation Technique	Key Mechanism	Remarks
P43 [cite]	GNN	Learns a parameterized explanation model that generalizes across instances. Improves scalability.	Uses probabilistic edge selection to approximate explanation distributions. Efficient for large graphs.	Scalable alternative to GNNExplainer.
P44 [cite]	GNN	Does not use explicit XAI but improves interpretability via learned representations. Captures structural similarity.	Uses contrastive learning between augmented graph views. Enhances feature learning without labels.	Useful for pretraining molecular models.
P45 [cite]	GNN	Uses attention weights to highlight important atoms and bonds. Provides implicit interpretability.	Self-supervised learning over large molecular graphs. Captures contextual chemical information.	Strong pretraining framework.
P46 [cite]	GNN (Attention)	Uses attention mechanisms to assign importance to molecular substructures. Enables interpretable predictions.	Graph attention layers focus on relevant nodes and edges. Learns weighted molecular representations.	Effective for toxicity prediction tasks.
P47 [cite]	GNN + CNN	Uses attention-based interpretation for drug-target interactions. Highlights binding regions.	Combines graph features with sequence-based CNN features. Learns joint representations.	Extends well to ADMET tasks.
P48 [cite]	XGBoost	Uses SHAP for feature importance analysis in tabular molecular descriptors. Enables global interpretability.	Gradient boosting over decision trees captures nonlinear relationships. Handles structured data effectively.	Strong baseline for descriptor-based models.
P49 [cite]	Random Forest	Uses SHAP/LIME to identify key molecular descriptors. Provides feature-level explanations.	Ensemble of decision trees reduces variance and improves robustness. Works well with limited data.	Reliable classical approach.
P50 [cite]	Deep Neural Network	Uses gradient-based attribution methods to interpret toxicity predictions. Highlights feature relevance.	Multi-layer neural networks learn complex nonlinear relationships. Suitable for large datasets.	High predictive performance but less interpretable.
P51 [cite]	Deep Neural Network	Applies Integrated Gradients to attribute feature contributions in chemical models. Enhances interpretability.	Uses gradient-based backpropagation to compute feature importance. Works across deep architectures.	Improves transparency of DL models.
P52 [cite]	QSAR models	Uses SHAP to analyze feature contributions in bioactivity prediction. Identifies relevant molecular fragments.	Applies feature attribution on classical ML models. Provides interpretable QSAR insights.	Important for medicinal chemistry insights.
P53 [cite]	ML models	Compares SHAP and LIME explanations across QSAR models. Evaluates consistency of explanations.	Uses perturbation-based and game-theoretic methods. Benchmarks interpretability.	Highlights reliability issues in XAI.
P54 [cite]	GNN	Uses counterfactual explanations to identify minimal structural changes. Provides actionable insights.	Applies genetic algorithms to modify molecular graphs. Searches for alternative predictions.	Useful for molecule optimization.
P55 [cite]	VAE/RL	Uses latent space interpretation to understand molecule generation. Links structure and properties.	Combines variational autoencoders with reinforcement learning. Optimizes molecular design.	Important for generative chemistry.

Table 2. *Cont.*

Paper	Model/Method	Explanation Technique	Key Mechanism	Remarks
P56 [cite]	Interpretable GNN	Integrates explainability directly into model architecture. Eliminates need for post-hoc methods.	Uses self-explaining nodes and attention mechanisms. Produces inherent explanations.	Improves trust in predictions.
P57 [cite]	Bayesian DL	Combines uncertainty estimation with SHAP explanations. Improves reliability.	Uses probabilistic neural networks to model uncertainty. Enhances robustness.	Useful in safety-critical domains.
P58 [cite]	Transformer	Uses attention weights to interpret SMILES representations. Captures token importance.	Pretrained language model for chemical sequences. Learns contextual embeddings.	Bridges NLP and chemistry.
P59 [cite]	Transformer	Uses attention-based explanations for molecular strings. Highlights key tokens.	Applies transformer architecture to SMILES encoding. Learns sequence dependencies.	Effective for property prediction.
P60 [cite]	Transformer	Uses self-attention to identify important tokens in molecular sequences.	Sequence modeling using transformer layers. Captures long-range dependencies.	Alternative to graph models.
P61 [cite]	3D Transformer	Uses spatial attention to interpret 3D molecular structures. Captures geometric features.	Learns 3D representations using transformer layers. Incorporates spatial information.	Advances structure-based prediction.
P62 [cite]	GNN	Uses geometric features for interpretability. Highlights spatial relationships.	Incorporates bond angles and distances into graph learning. Improves prediction of physical properties.	Useful for physical properties.
P63 [cite]	GNN	Uses directional message passing to interpret atomic interactions. Captures angular dependencies.	Models geometric relationships explicitly. Improves quantum property prediction.	Strong for physics-based tasks.
P64 [cite]	GNN	Uses continuous-filter convolutions for atomic interactions. Models interpretability based on spatial distances.	Learns representations for molecular energies. Widely used in chemistry.	Widely used in chemistry.
P65 [cite]	Neural Network	Uses physically inspired modeling to interpret interactions. Predicts energy surfaces.	Combines deep learning with physics constraints. Improves accuracy.	Suitable for quantum tasks.
P66 [cite]	Neural Network	Uses atom-centered features to explain molecular energies. Provides interpretable outputs.	Trained on large quantum datasets. Captures energy contributions.	Used in computational chemistry.
P67 [cite]	Transformer	Uses attention over graph structure to explain predictions. Captures global dependencies.	Combines graph encoding with transformer architecture. Improves performance.	Competes with GNNs.
P68 [cite]	Transformer	Uses generative modeling with interpretable token attention. Explains molecule generation.	Applies GPT-style architecture to molecules. Generates valid structures.	Emerging generative method.
P69 [cite]	ML/DL	Uses feature importance and model analysis tools. Provides basic interpretability.	Framework integrating multiple ML models. Supports drug discovery workflows.	Widely used toolkit.

Table 2. *Cont.*

Paper	Model/Method	Explanation Technique	Key Mechanism	Remarks
P70 [cite]	DL	Uses neural network interpretability tools. Enables model transparency.	Deep learning framework for chemical tasks. Supports modular modeling.	Flexible platform.
P71 [cite]	RL	Uses reward-based interpretation to guide molecule generation. Explains optimization.	Combines Monte Carlo tree search with RL. Optimizes molecular properties.	Effective for drug design.
P72 [cite]	VAE	Uses interpretable graph decomposition. Explains molecule generation process.	Builds molecules via tree structures. Ensures chemical validity.	Important generative model.
P73 [cite]	Flow-based GNN	Uses probabilistic modeling for explainable molecule generation. Captures distributions.	Autoregressive flow model for graphs. Generates molecules stepwise.	High-quality generation.
P74 [cite]	DL	Uses attention mechanisms for interpretability. Explains drug-target interactions.	Combines multiple encoders for drugs and proteins. Learns joint features.	Versatile framework.
P75 [cite]	CNN/RNN	Uses sequence attention to interpret binding interactions. Highlights important residues.	Combines CNN and RNN for sequence modeling. Predicts affinities.	Strong DTI baseline.
P76 [cite]	DL	Uses deep learning to model drug-target interactions. Provides feature-level insights.	Combines molecular and protein features. Improves prediction accuracy.	Early DL DTI model.
P77 [cite]	Deep NN	Uses shared representations for interpretability across endpoints. Identifies common patterns.	Multi-task learning modeling. Improves generalization.	Efficient ADMET framework.
P78 [cite]	Ensemble ML	Uses feature importance analysis across models. Provides global interpretability.	Combines multiple ML models. Improves robustness.	Comprehensive platform.
P79 [cite]	ML models	Uses feature attribution to interpret toxicity pathways. Links features to outcomes.	Trains on high-throughput screening data. Captures biological effects.	Large-scale toxicity modeling.
P80 [cite]	ML/DL	Uses interpretability tools to identify clinical risk factors. Explains predictions.	Models clinical toxicity endpoints. Uses benchmark datasets.	Clinically relevant.
P81 [cite]	ML/DL	Uses feature importance to interpret solubility predictions. Identifies key descriptors.	Predicts aqueous solubility. Uses regression models.	Simple benchmark task.
P82 [cite]	ML/DL	Uses feature attribution to explain hydration energy predictions.	Regression models trained on thermodynamic data.	Important for physical chemistry.
P83 [cite]	GNN	Uses attention to highlight features affecting BBB permeability.	Graph-based prediction of brain penetration.	Critical for CNS drugs.
P84 [cite]	ML/DL	Uses feature importance to explain side-effect predictions.	Models adverse drug reactions.	Important safety application.
P85 [cite]	GNN/ML	Uses attention and feature attribution to interpret binding affinity.	Learns drug-protein interactions.	Key for DTI tasks.

Table 2. *Cont.*

Paper	Model/Method	Explanation Technique	Key Mechanism	Remarks
P86 [cite]	GNN/CNN	Uses spatial attention to interpret binding interactions.	Models 3D protein-ligand structures.	Structural prediction.
P87 [cite]	ML	Uses feature importance to explain bioactivity predictions.	Uses large-scale screening data.	Broad applicability.
P88 [cite]	VAE/GAN	Uses latent representations to explain molecule generation.	Learns chemical distributions.	Drug design use.

Table 3. Summary of datasets and applications.

Paper	Classifier	Dataset	Application Domain
P89 [cite]	D-MPNN (GNN)	MoleculeNet (benchmark ADMET datasets), Tox21 (multi-endpoint toxicity dataset)	ADMET prediction
P90 [cite]	Self-supervised GNN	MoleculeNet (molecular benchmark datasets)	Molecular property prediction
P91 [cite]	Pretrained GNN	ChEMBL (large-scale bioactivity database), ZINC (drug-like compound library)	Drug property prediction
P92 [cite]	GNN with attention	MoleculeNet (benchmark ADMET datasets)	Toxicity prediction
P93 [cite]	GNN + CNN	BindingDB (protein-ligand binding dataset)	Drug-target interaction
P94 [cite]	Deep Neural Network	Tox21 (toxicity classification benchmark)	Toxicity prediction
P95 [cite]	XGBoost	Descriptor datasets (molecular feature vectors)	ADMET prediction
P96 [cite]	Random Forest	ChEMBL (bioactivity data repository)	Bioactivity prediction
P97 [cite]	RF / SVM	ChEMBL (bioactivity dataset for QSAR)	QSAR modeling
P98 [cite]	Deep Neural Network	Molecular datasets (various ADMET benchmarks)	Property prediction
P99 [cite]	GNN	DTI datasets (drug-target interaction benchmarks)	Drug-target prediction
P100 [cite]	Interpretable GNN	MoleculeNet (multi-task ADMET datasets)	ADMET prediction
P101 [cite]	Bayesian Neural Network	Molecular datasets (toxicity-focused datasets)	Toxicity prediction
P102 [cite]	GNN	MoleculeNet (molecular property datasets)	Counterfactual explanation
P103 [cite]	VAE/RL	ZINC (chemical compound database)	Molecule generation
P104 [cite]	Transformer	SMILES datasets (molecular string representations)	Representation learning
P105 [cite]	Transformer	PubChem (large chemical database)	Property prediction
P106 [cite]	Ensemble ML	Multiple ADMET datasets (integrated benchmarks)	Multi-endpoint ADMET
P107 [cite]	ML models	ToxCast (high-throughput toxicity screening)	Toxicity screening
P108 [cite]	ML/DL	ClinTox (clinical toxicity dataset)	Clinical toxicity prediction
P109 [cite]	ML/DL	ESOL (aqueous solubility dataset)	Solubility prediction
P110 [cite]	ML/DL	FreeSolv (hydration free energy dataset)	Thermodynamic prediction
P111 [cite]	GNN/ML	Lipophilicity (logD dataset)	Pharmacokinetics prediction
P112 [cite]	GNN	BBBP (blood-brain barrier dataset)	BBB permeability prediction
P113 [cite]	ML/DL	SIDER (drug side-effect dataset)	Side-effect prediction

Table 3. Cont.

Paper	Classifier	Dataset	Application Domain
P114 [cite]	GNN/ML	BindingDB (binding affinity dataset)	Drug-target binding
P115 [cite]	GNN/CNN	PDBbind (protein-ligand complexes)	Binding affinity prediction
P116 [cite]	ML	PubChem (bioassay screening data)	Bioactivity prediction
P117 [cite]	VAE / GAN	ZINC (compound library)	Drug design
P118 [cite]	Deep Neural Network	MoleculeNet (multi-task ADMET datasets)	Multi-endpoint ADMET
P119 [cite]	ML/DL	MoleculeNet (benchmark suite)	Drug discovery prediction
P120 [cite]	Deep Learning	ChEMBL (bioactivity dataset)	Molecular property prediction
P121 [cite]	Reinforcement Learning	ZINC (compound dataset)	Molecule generation
P122 [cite]	VAE	ZINC (molecular dataset)	Molecular graph generation
P123 [cite]	Flow-based GNN	ZINC (chemical dataset)	Molecule design
P124 [cite]	DL framework	BindingDB (interaction dataset)	Drug-target prediction
P125 [cite]	CNN/RNN	BindingDB (binding dataset)	Binding affinity prediction
P126 [cite]	Deep Learning	DrugBank (drug database)	DTI prediction
P127 [cite]	Transformer	PubChem (chemical database)	Property prediction
P128 [cite]	Transformer	SMILES datasets (chemical strings)	Representation learning
P129 [cite]	3D Transformer	Molecular datasets (3D conformations)	Structure-based prediction
P130 [cite]	GNN	Molecular graphs (3D features)	Property prediction
P131 [cite]	GNN	QM9 (quantum chemistry dataset)	Molecular property prediction
P132 [cite]	GNN	QM9 (quantum dataset)	Energy/property prediction
P133 [cite]	Neural Network	QM9 (quantum dataset)	Energy prediction
P134 [cite]	Neural Network	QM datasets (molecular energy)	Force field prediction
P135 [cite]	DL	Molecular datasets (various)	Drug discovery prediction
P136 [cite]	Reinforcement Learning	ZINC (compound dataset)	Molecule optimization
P137 [cite]	Transformer	Molecular graphs (ADMET datasets)	Property prediction
P138 [cite]	Transformer	ZINC (compound library)	Molecule generation

Table 4. Strengths and limitations of various classifiers.

Classifier	Type	Usage in Drug Discovery	Strengths	Limitations
Graph Neural Networks (GNN)	Deep Learning	Molecular property prediction, ADMET modeling, toxicity prediction	Captures molecular graph structure and relationships effectively	Computationally expensive, requires large datasets
Message Passing Neural Networks (MPNN)	Deep Learning	ADMET prediction, quantum property modeling	Learns atom-level interactions through message passing	Limited interpretability without XAI methods
Graph Attention Networks (GAT)	Deep Learning	Molecular prediction, toxicity modeling	Highlights important atoms and bonds via attention	Can overfit on small datasets

Table 4. *Cont.*

Classifier	Type	Usage in Drug Discovery	Strengths	Limitations
Transformer Models (ChemBERTa, MolBERT)	Deep Learning	SMILES-based prediction, molecular representation learning	Captures long-range dependencies in sequences	Requires large-scale pretraining
3D Transformers (Uni-Mol)	Deep Learning	Structure-based drug design, binding prediction	Incorporates spatial and geometric information	High computational cost
Random Forest (RF)	Machine Learning	QSAR modeling, bioactivity prediction	Robust, interpretable with SHAP, works on small datasets	Limited ability to capture complex relationships
XGBoost	Machine Learning	ADMET prediction, toxicity modeling	High accuracy on tabular data, efficient training	Less interpretable than simple models
Support Vector Machine (SVM)	Machine Learning	QSAR modeling, classification tasks	Effective in high-dimensional feature spaces	Not scalable to very large datasets
Deep Neural Networks (DNN)	Deep Learning	Toxicity prediction, multi-task ADMET	Captures complex nonlinear relationships	Black-box nature, requires large data
Convolutional Neural Networks (CNN)	Deep Learning	Protein sequence modeling, drug-target prediction	Effective for sequence-based learning	Limited for graph-based molecular data
Recurrent Neural Networks (RNN)	Deep Learning	SMILES modeling, sequence prediction	Handles sequential dependencies	Vanishing gradient issues
Variational Autoencoders (VAE)	Deep Learning	Molecule generation, latent space modeling	Generates diverse chemical structures	May produce invalid molecules
Generative Adversarial Networks (GAN)	Deep Learning	Drug design, molecule generation	Produces realistic synthetic molecules	Training instability
Reinforcement Learning (RL) Models	Machine Learning	Molecule optimization, drug design	Enables goal-directed molecule generation	Requires careful reward design
Bayesian Neural Networks (BNN)	Probabilistic DL	Toxicity prediction with uncertainty estimation	Provides uncertainty and reliability estimation	Computationally intensive